

Hybrid Systems Architecture Comparison

Critical Issue: Hybrid extrapolation 60% R^2 vs Pure LLM 100% R^2

System 1: Improved Hybrid

✅ FIXES Extrapolation Weakness

LLM Engine

Formula discovery

NN Engine

Deep learning

Pattern Recognition & Confidence

Few-shot examples • Confidence scoring

⚠️ EXTRAPOLATION-AWARE DECISION ⚠️

Detects out-of-range data

Prefers LLM for extrapolation

`if is_extrap and llm_r2 > 0.90: use LLM`

Ensemble Output

Optimized adaptive weights

✅ Target: 95-100% Extrapolation R^2

(vs 60% baseline)

System 2: Symbolic Discovery

❌ Does NOT Fix Extrapolation

Symbolic Regression Engine

PySR • gplearn • Formula discovery

4-Layer Validation System

✓

Layer 1: Symbolic Validation

Syntax checks

✓

Layer 2: Dimensional Analysis

Unit consistency

✓

Layer 3: Domain Knowledge

DeFi rules

✓

Layer 4: Numerical Validation

Edge cases

LLM Interpretation (Optional)

Formula naming only

⚠️ Target: 85+ validation score

(NOT extrapolation R^2)

⚠️ Key: System 1 decides WHEN to use LLM vs NN • System 2 only validates ONE formula